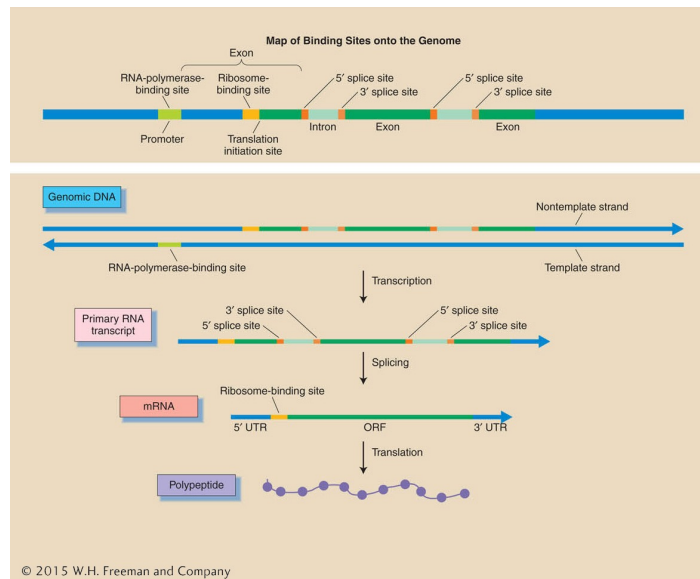


RNA Sequencing (RNA-Seq)

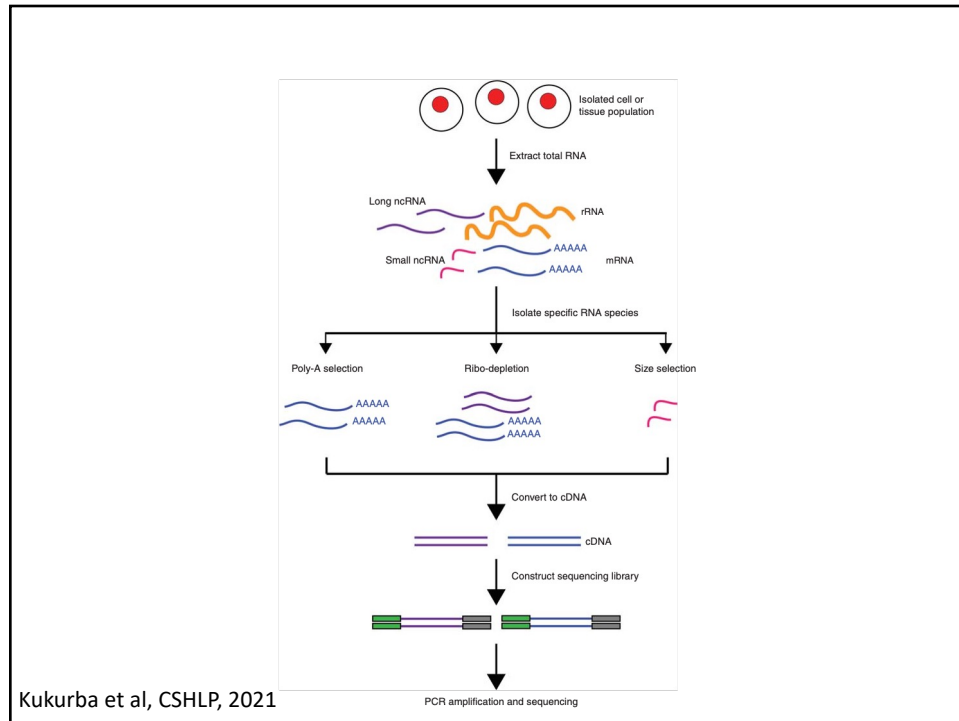
- Used to obtain gene expression information (not all genes are expressed all the time!)
 - Which genes are expressed in which conditions?
 - Which genes are expressed in which cells?
 - Which genes are expressed in which tissues?
 - Detects isoforms

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Genome Annotation

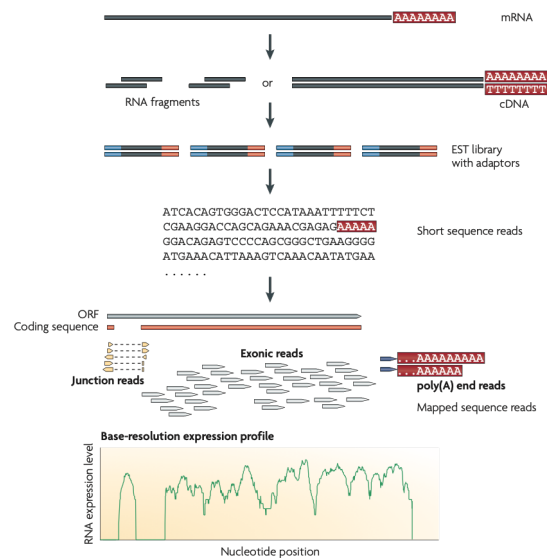


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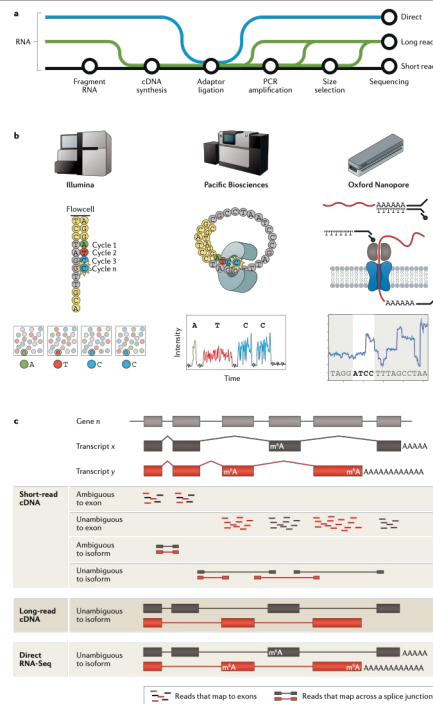
129

Procedure of RNA-Seq



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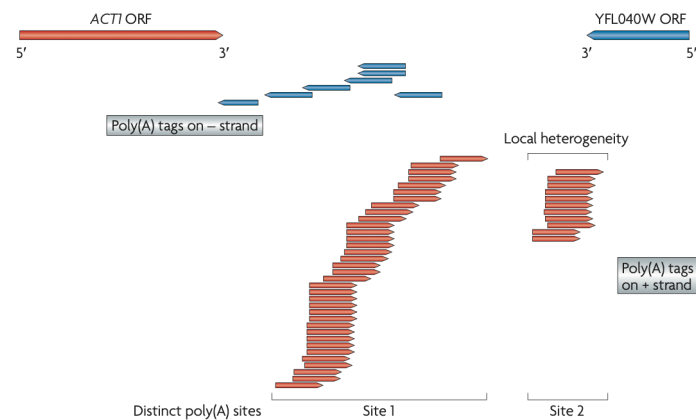
Procedure of RNA-Seq



Stark et al, Nat. Rev. Gen. 2019

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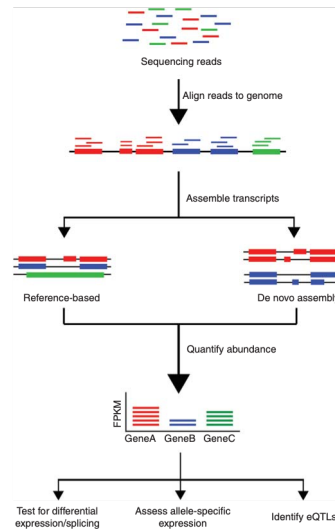
A beautiful example in *Saccharomyces cerevisiae*



Wang et al, Nat. Rev. Gen., 2009

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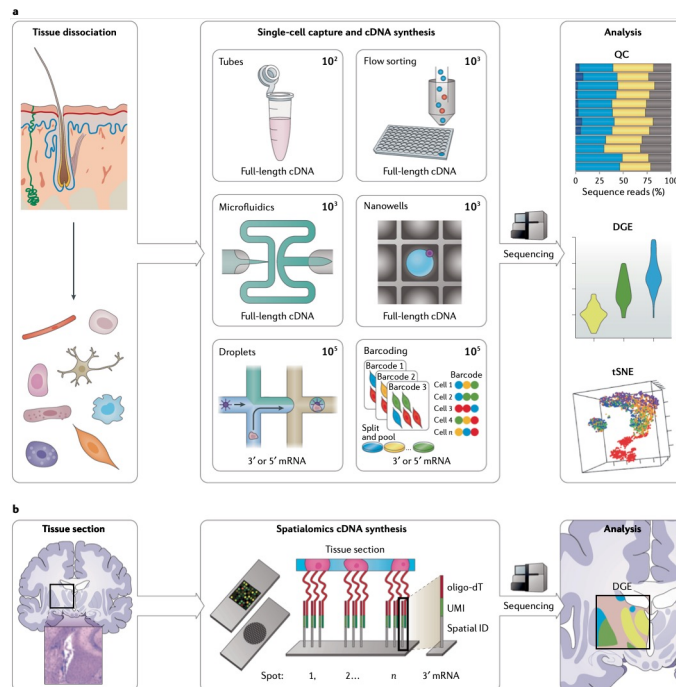
RNA-Seq Data Analysis



Kukurba et al, CSHLP, 2021

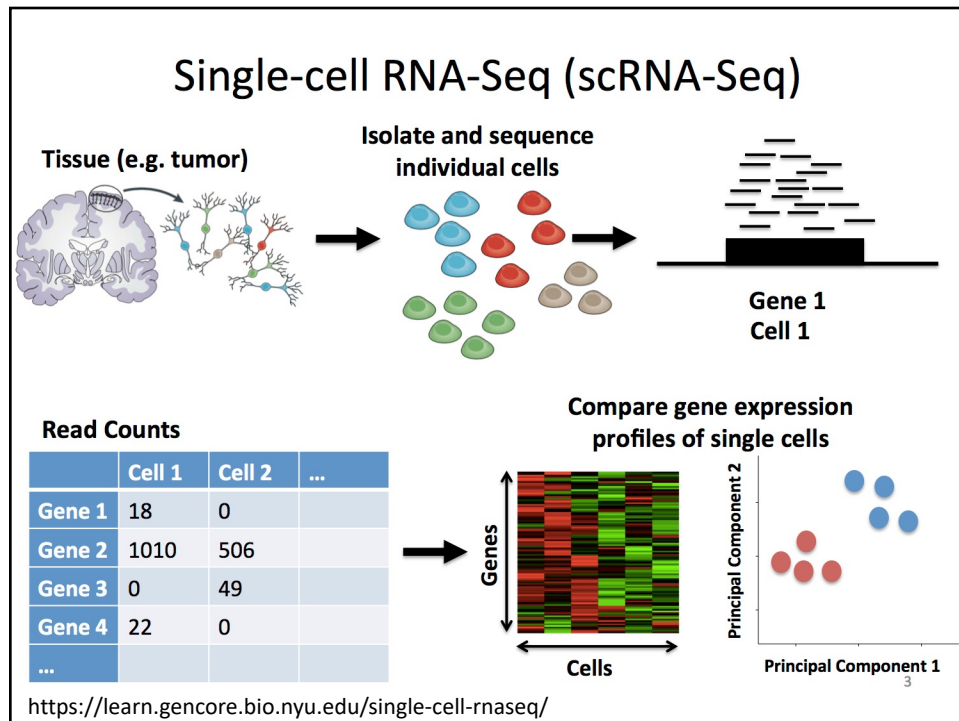
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Single-cell RNA-Seq

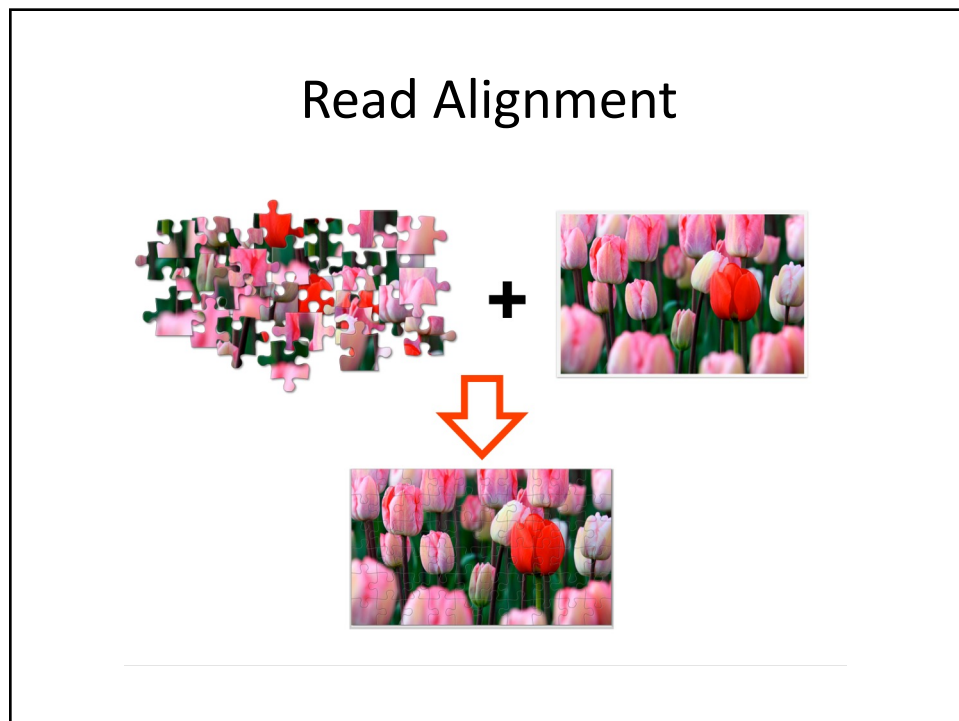


Stark et al, Nat. Rev. Gen. 2019

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Why is it important?

- Reference-guided genome assemblies
- Identification of Single Nucleotide Polymorphisms (SNPs, mutations)
- RNA-Seq analysis

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Why is it hard?

Read

>SRR769401.3.1 3 length=101
TAGCTTTCAGTGCCTGAGAGTATTTCCCATTTTCAGTAAATCCGAAAGCTTGTAGCAAAATAAGTGGCTTGAGCATCTGTTATTCTGGAATAATATC

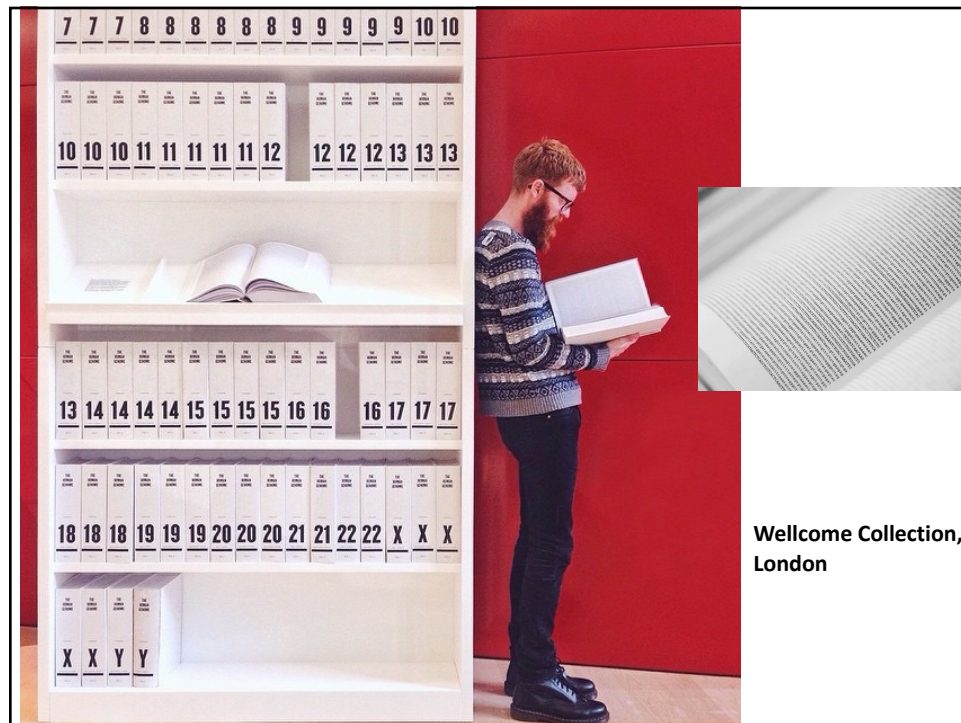
X 200,000,000 TIMES

Reference

>Homo sapiens, complete genome
GATCACAGGTCTATCACCTATTAACTCACTACGGGAGCTCTCCATGCATTGGTATTTTCGTCTGGGG
GTATGCACGCGATAGCATTGCGAGACGCTGGAGCCGGAGCACCTATGTCGAGTATCTGCTTTGATT
CTGCTCATCTATTATTATCGACCTACGTTCAATATTACAGGCGAACATACTACTAAAGTGTGTA
ATTAATTAATGCTTTAGGACATAATAACAATTGAATGTCTGCACAGCCACTTCCACACAGACATC
ATAACAAAAATTTCCACAAACCCCTCCCTCCCTCTGGCCACAGCACTAAACACATCTCTGCA
AACCCAAAAACAAAGAACCTAACACAGCTAACAGATTCAAAATTTATCTTTGGCGGTATGCA
TTTAAACAGTCACCCCACTAACACATTATTTCCCTCCCACTCCCACTACTACTATCTCATCAATA
CAACCCCGCCCATCTACCCAGCACACACACCGCTGCTAACCCCATACCCGAACCAACCAACCC
AAAGACACCCCCACAGTTTATGTAGCTTACCTCCTCAAGCAATACACTGAAATGTTTAGACGGGCTC
ACATCACCCATAAACAAATAGGTTTGTCTAGCCTTTCTATTAGCTCTTAGTAAGATTACACATGCAA
GCATCCCGTTCCAGTGAGTTACCCCTAAATCACACGATCAAAAGGAACAAGCATCAAGCAGCGAG
AATGACGCTCAAAACGCTTAGCTAGCCACACCCACGGGAAACAGCAGTGAATTAACCTTTAGCAATA
ACGAAAGTTTAACTAAGCTATACTAACCCAGGTTGGTCAATTTCTGTCCAGCACCGGTCACACGA
TTAACCAAGTCAATAGAACCGGCTAAAGAGTGTGTTAGATCAACCCCTCCCAATAAAGCTAAAC
CACCTGAGTTGTAATACTCCAGTTGACACAAATAGACTACGAAAGTGGCTTTAATATCTGAACAC
ACAATAGCTAAGACCAAACTGGGATTAGATACCCCACTATGCTTAGCCCTAAACCTCAACAGTTAAATC
AACAAAACTGCTGCCAGAACACTACGAGCCACAGCTTAAACTCAAGGACCTGGCGGTGCTTATATC
CTCTAGAGGAGCTGTTCTGTAATCGATAAACCCGATCAACCTCACACCTCTTGCTCAGCCATATA
CCGCTCATCTTCAGCAACCTGATGAAGGCTACAAAGTAAGCGCAAGTACCCAGTAAAGACGTTAGGTC

X 3,000,000 TIMES

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Wellcome Collection,
London

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Strings

- A *string* S is a finite ordered list of characters, where characters are drawn from an alphabet Σ .
- Nucleic acid alphabet: $\{ A, C, G, T \}$
- Amino acid alphabet: $\{ A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y, V \}$
- ϵ is the empty string.
- S is a *substring* of T if there exist (possibly empty) strings u and v such that $T = uSv$
- *Subsequence* is similar except the characters need not be consecutive.
 - “cat” is a substring and a subsequence of “concatenate”
 - “cant” is a subsequence of “concatenate”, but not a substring

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1. Naïve exact matching

At what offsets does *pattern P* occur within *text T*?

P: word

T: There would have been a time for such a word

P: word

T: There would have been a time for such a word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word

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1. Naïve exact matching

- More character comparisons are done when possible matches exist

P: word

T: There would have been a time for such a word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word
 word word word word word word word word word word

40 mismatches + 6 matches = 46 character comparisons

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2. Boyer-Moore Exact Matching

Exact matching: better naïve algorithm

P: word

T: There would have been a time for such a word

-----word----->

In the naïve mismatch algorithm, you would continue to explore the next alignment

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2. Boyer-Moore Exact Matching

Learn from character comparisons to skip pointless alignments

Try alignments in left-to-right order, and try character comparisons in right-to-left order

P: word

T: There would have been a time for such a word

-----word----->

Boyer, RS and Moore, JS. "A fast string searching algorithm."
Communications of the ACM 20.10 (1977): 762-772.

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Boyer-Moore: Bad character rule

Upon mismatch, skip alignments until (a) mismatch becomes a match, or (b) P moves past mismatched character

Step 1: T : G C T T **C** T G C T A C C T T T T G C G C G C G C G G A A
 P : C C T T **T** T G C



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Boyer-Moore: Good suffix rule

Let t = substring matched by inner loop; skip until (a) there are no mismatches between P and t or (b) P moves past t

Step 1: T : C G T G C C **T A C** T T A C T T A C T T A C G C G A A
 P : C T T A C **T T A C**



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Boyer-Moore: Putting it together

Use bad character or good suffix rule, *whichever skips more*

Step 1:
 T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
 P: G T A G C G G C G bc: 6, gs: 0 bad character

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2. Boyer-Moore Exact Matching

11 characters of T we ignored
 ■■■■■■■■ ■■■■

Step 1:
 T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
 P: G T A G C G G C G

Step 2:
 T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
 P: G T A G C G C G

Step 3:
 T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
 P: G T A G C G G C G

Step 4:
 T: G T T A T A G C T G A T C G C G G C G T A G C G G C G A A
 P: G T A G C G G C G

■■■■■■■ ■■ ■■■■■■■■
 Skipped 15 alignments

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